Global Development Indicators: BI Analysis Report

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1. Executive Summary

Leadership needs a clear view of where health and economic development lag behind, and where deeper policy review is needed. Using a cleaned Gapminder country-year dataset (142 countries, 1952-2007) in Tableau, the dashboard shows that average life expectancy rose from 49.1 years in 1952 to 67.0 years in 2007, with Asia improving most (+24.4 years). Africa, however, still had the lowest life expectancy in 2007 (54.8 years) and contains 24 of the 25 countries in the High Priority band. The top recommendation is to prioritise Sub-Saharan Africa for deeper policy review, starting with these 24 High Priority countries.

2. Methodology

- **Data:** Dataset.csv (6,856 rows, 15 columns) in long format: one row per country, year and indicator. Source: Gapminder public development indicators, accessed through Plotly sample data.
- **Tool:** Tableau Desktop. The raw CSV was not edited; all cleaning was done inside Tableau and is documented on dashboard page 3.
- **Text cleaning:** calculated fields using TRIM and PROPER for Country (184 raw spellings -> 142 countries), Continent (10 -> 5), Priority Band (6 -> 3) and Income Context (8 -> 4, hyphens also standardised; 55 blanks labelled 'Not recorded').
- **Duplicates:** 40 duplicate rows were identical except for record_id and source_note, so a check excluding only record_id finds none. They were flagged with CONTAINS([Source Note], 'DUPLICATE') and removed with a data source filter, leaving 6,816 rows (142 countries x 12 years x 4 indicators).
- **Missing values:** 35 blank indicator values (0.5%) were counted by indicator and left blank rather than estimated; Tableau ignores them in averages and sums.
- **Data types:** Year as a discrete dimension, Indicator Value as a decimal measure, ISO alpha code given the Country/Region geographic role for the map.
- **KPIs and filters:** one calculated field per indicator code so units are never mixed; AVG for life expectancy and GDP per capita, SUM for population in a single year only. Filters: Year (default 2007), Continent, Priority Band, Country and a single-value Indicator selector.

3. Dashboard Overview

- **Page 1 - Executive Summary:** KPI cards (2007), life-expectancy trend by continent, priority-band map, population bar chart, GDP vs life-expectancy scatter, and insight/recommendation text. Supports the decision of where leadership should focus first.

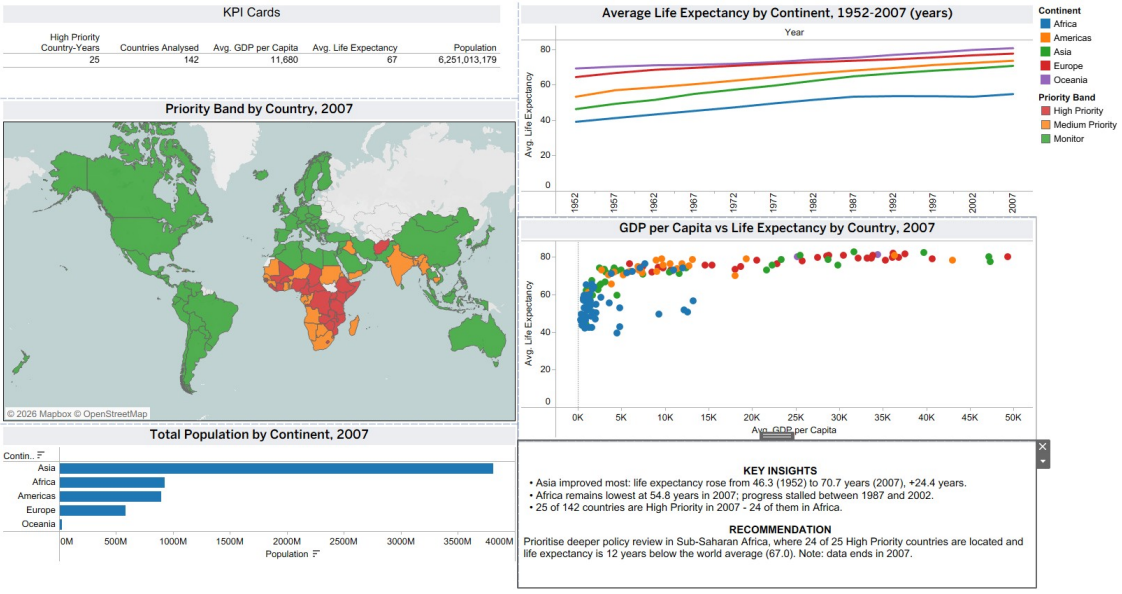


Figure 1. Executive Summary dashboard (all continents, 2007).

- **Page 2 - Country Deep Dive:** ranked country bar chart and a trend line for the indicator chosen in the dropdown, with continent and country filters. Bars are coloured by priority band so High Priority countries stand out. Supports analysts comparing specific countries and regions.

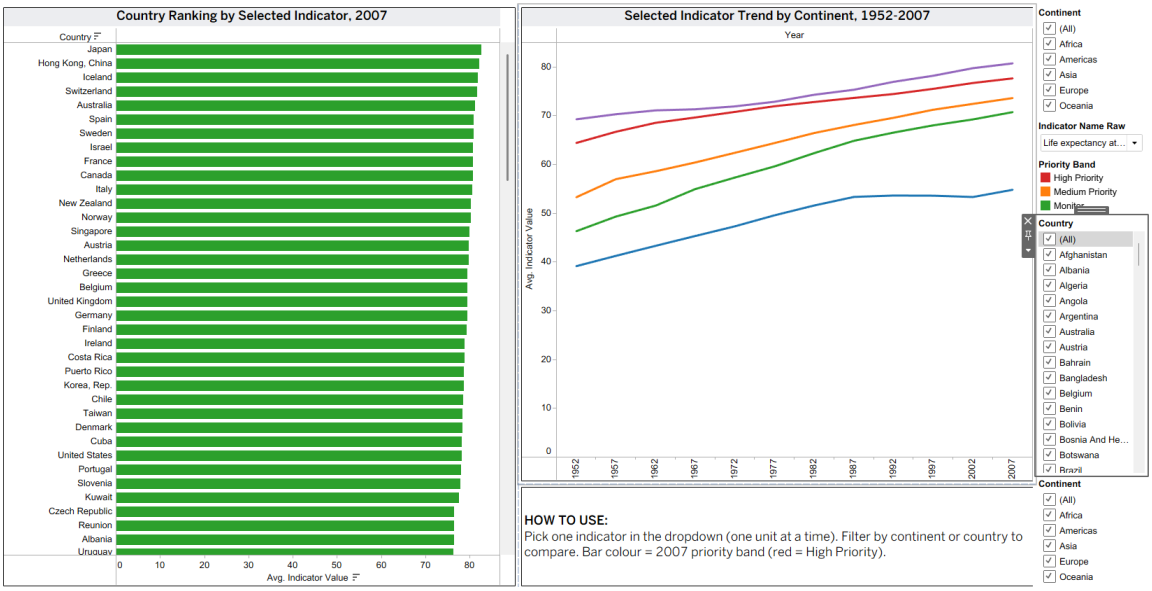


Figure 2. Country Deep Dive (life expectancy selected).

- **Page 3 - Data Quality & Method:** summary of the cleaned dataset (6,816 rows, 142 countries, 12 years, 4 indicators, 35 missing values), a missing-values bar chart by indicator, a raw-vs-clean label check table, and notes on cleaning decisions and limitations, so the work can be audited.

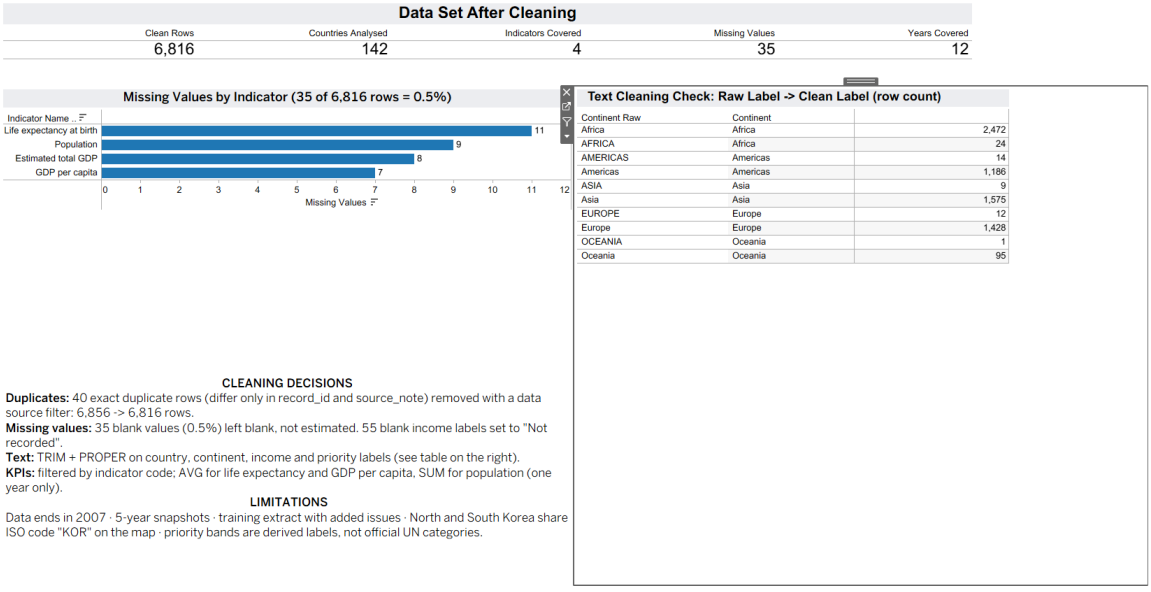


Figure 3. Data Quality & Method Notes.

4. Key Insights

- **Insight 1 - Asia improved most.** Average life expectancy in Asia rose from 46.3 years (1952) to 70.7 years (2007), +24.4 years, the largest gain of any region. The Americas gained +20.3, Africa +15.7, Europe +13.2 and Oceania +11.5 years.
- **Insight 2 - Africa's progress stalled.** Africa's average life expectancy was 53.3 years in 1987 and still 53.3 in 2002, before rising to 54.8 in 2007. It remains 12.2 years below the 2007 average of all countries (67.0) and 22.8 years below Europe (77.6).
- **Insight 3 - High Priority contexts are concentrated in Africa.** In 2007, 25 of 142 countries were High Priority: 24 in Africa plus Afghanistan. Filtering the dashboard to Africa shows 52 countries, an average GDP per capita of $3,089 and average life expectancy of 55 years. The number of High Priority countries fell from 79 in 1952 to 25 in 2007, so progress is real but uneven.
- **Insight 4 - Income matters, but not alone.** The scatter plot shows that higher GDP per capita goes with higher life expectancy, levelling off above about $10,000. Yet some middle-income African countries have low life expectancy, e.g. Botswana ($12,570; 50.7 years) and South Africa ($9,270; 49.3 years).
- **Insight 5 - Scale is in Asia.** Asia holds 3.81 billion of the 6.25 billion people analysed in 2007 (61%), followed by Africa with 930 million, so changes in Asia have the largest absolute planning impact.

5. Recommendations

- **1. Prioritise Sub-Saharan Africa for deeper policy review.** Evidence: 24 of 25 High Priority countries are in Africa (priority map, KPI cards filtered to Africa). Start with the 24 countries in red on the map.
- **2. Investigate why life expectancy stalled in Africa between 1987 and 2002.** Evidence: the Africa line on the trend chart is flat at about 53 years for 15 years. A review should look for the causes behind the plateau before designing new programmes.
- **3. Look beyond income in middle-income countries with low life expectancy.** Evidence: the scatter plot shows countries such as Botswana and South Africa with GDP per capita above $9,000 but life expectancy near 50. Health-system and health-specific indicators should be reviewed for these countries, not only economic growth.
- These recommendations are based on 2007 data and derived priority labels (see Limitations); they should be confirmed with current data before final policy decisions.

6. Limitations

- **Data age:** the dataset ends in 2007, so it does not reflect current conditions.
- **Time coverage:** values are 5-year snapshots (1952, 1957, ... 2007), so changes between those years are not visible.
- **Missing values:** 35 indicator values (0.5%) are blank and 55 income-context labels are 'Not recorded'; averages use only available values.
- **Training modifications:** the extract contains deliberately added quality issues and duplicate rows; priority bands and income contexts are derived labels, not official UN classifications.
- **Map limitation:** North Korea and South Korea share the ISO code 'KOR' in the source, so the map shows them as one location.
- **Assumptions:** averages are simple (unweighted) country averages, so a small country counts as much as a large one.

7. Appendix

- **Calculated fields:** Country, Continent, Priority Band = PROPER(TRIM(raw field)); Income Context = IF ISNULL OR blank THEN 'Not recorded' ELSE PROPER(REPLACE(TRIM(raw),'-',' ')); Duplicate Flag = CONTAINS([Source Note],'DUPLICATE'); Missing Value Flag = ISNULL([Indicator Value]).
- **KPI fields:** Life Expectancy / Population / GDP per Capita = IF [Indicator Code] = code THEN [Indicator Value] END; Countries Analysed = COUNTD([Country]); High Priority Country-Years = COUNTD(IF [Priority Band]='High Priority' THEN [Country]+'-'+STR([Year]) END).
- **2007 KPI results:** average life expectancy 67.0 years; total population 6,251,013,179; average GDP per capita $11,680; 142 countries; 25 High Priority countries.